Linear and Planar Densities in the [111] Direction and (111) Plane of Fe (BCC) and Ni (FCC)

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> *6. Calculate the linear density in the* $[111]$ *direction and the planar density in the* (111) *plane for:* ***a)*** *Iron BCC (Solution:* $\rho_{[111]} = 2/(\ \ 3a)$, $\rho_{(111)} = 1/(a^2\ \ 3)$*)* ***b)*** *Nickel FCC (Solution:* $\rho_{[111]} = 1/(a\ \ 3)$, $\rho_{(111)} = 4/(a^2\ \ 3)$*)*
>
> **Solution:** $\rho_{[111]} = 2/(\ \ 3a)$, $\rho_{(111)} = 1/(a^2\ \ 3)$

PHASE 1: HYPOTHESIS & PARAMETERS

Structural Models:

1. **Ferritic Iron (α-Fe):** Body-Centred Cubic (BCC) lattice, parameter a_{Fe}, $n = 2$ atoms/cell.
2. **Nickel (Ni):** Face-Centred Cubic (FCC) lattice, parameter a_{Ni}, $n = 4$ atoms/cell.

Comparison Parameters:

- Direction evaluated: $[111]$ (main cube diagonal, length $L_{[111]} = a\ \ 3$).
- Plane evaluated: (111) (plane that intersects the axes at $x = a, y = a, z = a$, bounding an equilateral triangle of edges $a\ \ 2$).

PHASE 2: FUNDAMENTAL PHYSICS & FORMULA SELECTION

Citing the lattice packing theory [Session 4 Slides 7-9]:

- **Linear Density:**

$$\rho_{[111]} = \frac{N_{\text{atoms centred on the diagonal}}}{a\ \ 3}$$

- **Planar Density:**

$$\rho_{(111)} = \frac{N_{\text{atoms contained in the triangle}}}{A_{\text{triangle}}}$$

where the triangle on the cell has edge $L = a\quad 2$ and area:

$$A_{\text{triangle}} = \frac{3}{4}L^2 = \frac{3}{4}(2a^2) = \frac{3}{2}a^2$$

PHASE 3: ANALYTICAL DERIVATION & CALCULATION STEP-BY-STEP

Part a: BCC Iron (α-Fe)

1. **Linear Density on** $[111]$**:**
2. In BCC, atoms are in contact along the main diagonal $[111]$.
3. The diagonal passes through the two opposite vertices and goes through the central body atom.
4. Atoms cut: $2 \times \frac{1}{2}(\text{vertices}) + 1(\text{centre}) = 2$ atoms.
5. Segment length: $L = a\quad 3$.

$$\rho_{[111]} = \frac{2}{a\quad 3} = \frac{\mathbf{2}}{\mathbf{3a}}$$

6. **Planar Density on** (111)**:**
7. The (111) plane contains only the 3 vertices $(a,0,0), (0,a,0), (0,0,a)$. Each vertex contributes an angle of 60° inside the equilateral triangle ($\frac{60^\circ}{360^\circ} = \frac{1}{6}$):

$$N_{\text{atoms}} = 3 \times \frac{1}{6} = \frac{1}{2}\ \text{atom}$$

- The central atom $(a/2, a/2, a/2)$ is **not contained in the** (111) **plane**, but lies at a perpendicular distance $d = \frac{a}{2\quad 3}$ behind it.
- Area of the triangle: $A = \frac{3}{2}a^2$.

$$\rho_{(111)} = \frac{1/2}{\frac{3}{2}a^2} = \frac{\mathbf{1}}{\mathbf{a^2\quad 3}}$$

Part b: FCC Nickel (Ni)

1. **Linear Density on** $[111]$**:**
2. In FCC, atoms do not touch along $[111]$; the cube diagonal is empty at its centre.
3. The diagonal cuts only the two atoms at the opposite vertices through their centres:

$$N_{\text{atoms}} = 2 \times \frac{1}{2} = 1 \text{ atom}$$

- Length: $L = a \quad 3$.

$$\rho_{[111]} = \frac{\mathbf{1}}{\mathbf{a} \quad \mathbf{3}}$$

4. **Planar Density on** (111)**:**
5. In FCC, the (111) plane is the plane of maximum packing (close-packed plane).
6. It contains the 3 vertices of the triangle (each contributes $\frac{1}{6}$) plus the 3 face centres located at the midpoints of the triangle sides (each contributes $\frac{1}{2}$ since it belongs to two adjacent cells):

$$N_{\text{atoms}} = 3 \times \left(\frac{1}{6}\right) + 3 \times \left(\frac{1}{2}\right) = \frac{1}{2} + \frac{3}{2} = 2 \text{ atoms}$$

- Area of the triangle: $A = \frac{\sqrt{3}}{2}a^2$.

$$\rho_{(111)} = \frac{2}{\frac{\sqrt{3}}{2}a^2} = \frac{\mathbf{4}}{\mathbf{a^2} \quad \mathbf{3}}$$

PHASE 4: PHYSICAL INTERPRETATION & VERIFICATION

Study notes written from the official course material (Problems T2_CrystStruct.pdf; Problems T2 defects.pdf).
Online version: subjects/aerospace-materials-1/problemas/topic-2-problems.html